Vibe *Check*

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CIS 2450 Big Data · Final Project

PROBLEM & VALUE PROPOSITION

Genre is too coarse. Co-listening is too cold-start. There's a gap.

Mood-based listening is how people actually consume music — "something for studying," "music to cry to," "a Saturday-night party mix" — but the tools that surface those tracks rely either on hand-curated playlists or on co-listening signals that take weeks to accumulate. New releases sit in a discovery dead zone.

WHAT'S MISSING IN PRACTICE

- **Genre is too coarse —** "pop" can mean a heartbreak ballad or a beach anthem; the same artist can hit both in one album.
- **Co-listening fails on cold start —** Spotify Daily Mixes and Apple's For-You feed both need weeks of stream data before a new release surfaces.
- **Manual mood labels don't scale —** editorial teams curate ~10k playlists; the catalog has 100M+ tracks.

WHAT VIBE CHECK DELIVERS

- **Content-only classification —** every track gets an archetype label from its features and lyrics alone. No listening data required.
- **Cold-start coverage —** can tag any new release the moment it's uploaded; populate mood playlists before any user has streamed it.
- **Production economics —** trains efficiently on a laptop, predicts in microseconds, no GPU. Pennies per million tracks.

DATA SOURCES

Three datasets, joined into one — and why each one matters

Spotify MPD

1M+ PLAYLISTS

Million Playlist Dataset gives us the playlist-to-track membership matrix. We score each playlist's name and description against an archetype-keyword dictionary, then propagate the strongest match down to its tracks. This is what generates our labels.

Spotify 1.2M Songs

14 AUDIO FEATURES

Per-track metadata: danceability, energy, valence, tempo, loudness, speechiness, acousticness, instrumentalness, liveness, key, mode, duration_ms, time_signature, year. These are the model's audio inputs.

Genius Lyrics

~5M LYRICS

Fuzzy-matched (RapidFuzz, threshold 85) on artist + title. Of the 32k labeled tracks, 71% had matched lyrics. Each matched track ran through VADER, j-hartmann emotion, and sentence-transformer scoring.

AFTER JOIN, DEDUPLICATION, IMPUTATION, AND ENCODING

32,112 tracks · **37** features · **6** archetypes · **71%** lyric-match rate

14 audio + 18 lyric (NLP-derived) + 5 categorical features. 29% of tracks are instrumentals or unmatched — handled with structural imputation and KNN imputation (k=5).

THE LABELS

Six listening archetypes — derived from playlist keywords

📚 Study / Focus

8,105 tracks · 25.2%

Lo-fi, ambient, instrumental

🎷 Jazz / Lounge

7,748 tracks · 24.1%

Smooth jazz, lounge, late-night

🌴 Summertime / Party

4,692 tracks · 14.6%

Beachy, upbeat, danceable

💔 Heartbreak / Sad

4,489 tracks · 14.0%

Sad, lonely, melancholic

🤠 Rustic / Nostalgia

3,592 tracks · 11.2%

Country, folk, Americana

🔥 Hype / Rap

3,486 tracks · 10.9%

High-energy rap and hip-hop

PIPELINE

Six stages from raw playlists to a 37-feature labeled dataset

01

Source

Spotify MPD
+ Spotify 1.2M
+ Genius

02

Label

Keyword-score playlists; tracks inherit archetype

03

Audio EDA

Distributions, outliers, feature correlations

04

Lyric NLP

VADER + emotion + sentence transformers

05

ML Prep

Train/test split, KNN impute, encode cats

06

Model

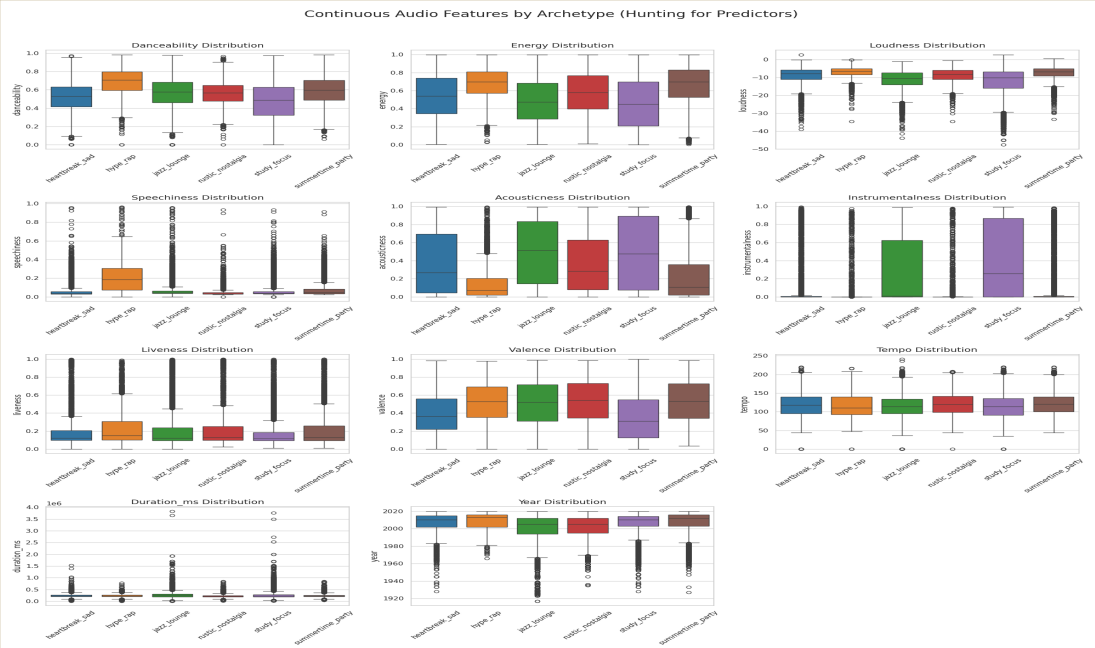
Decision tree, Radom Forest, Light GBM,

EVERY DECISION DOWNSTREAM IS GROUNDED IN AN UPSTREAM EDA RESULT

We chose tree-based ensembles after the audio-correlation matrix showed non-trivial multicollinearity. We built the lyric NLP pipeline because audio-only EDA showed Heartbreak/Sad and Summertime/Party occupying nearly identical audio space. We used class-balanced weights because the archetype distribution skews 2.3× toward Study/Focus.

EDA · FINDING 1 OF 4

Audio features genuinely separate the archetypes — but not all of them



KEY DATA POINT *Hype/Rap median speechiness: 0.30 · All other archetypes: <0.07. That's a 4× separation on a single feature.*

FINDING

Speechiness is a near-binary signal for Hype/Rap. Instrumentalness is a near-binary signal for Study/Focus. Energy and valence cluster archetypes into a high-arousal cluster vs. a low-arousal cluster.

WHY

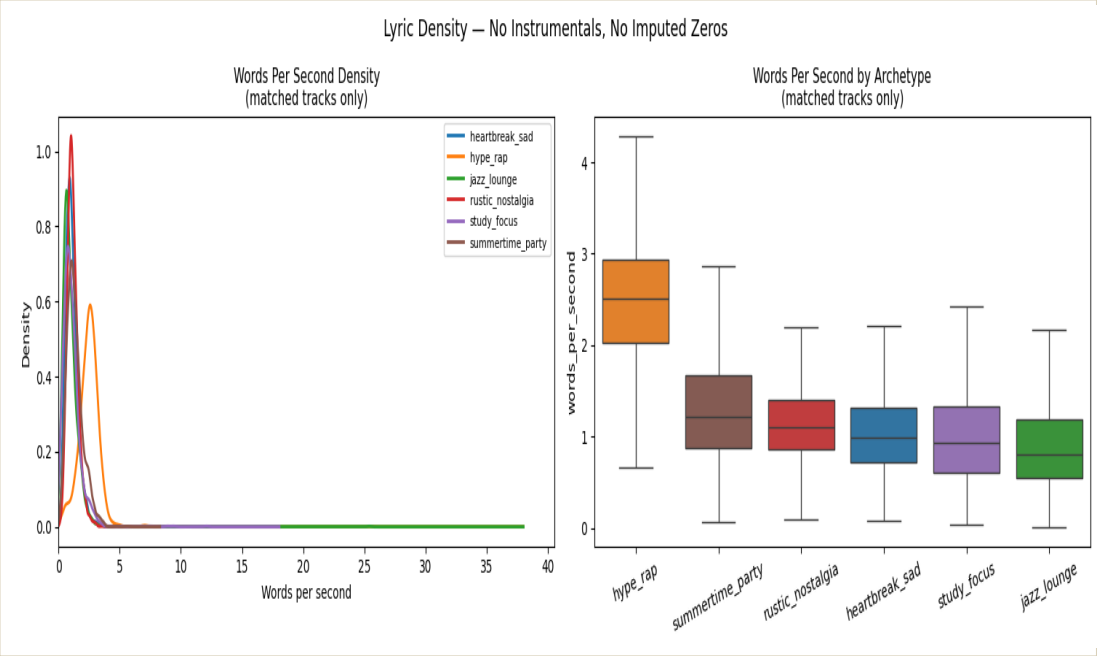
Spotify's speechiness measures spoken-word density; rap is structurally high-density vocal music. Instrumentalness flags tracks with no vocals — and lo-fi study music plus jazz standards are mostly instrumental.

WHAT IT MEANS FOR THE MODEL

Audio alone gets us to 0.573 macro F1 — solid. But Heartbreak/Sad and Summertime/Party still overlap heavily in audio space. That's the gap that motivated building the lyric pipeline.

EDA · FINDING 2 OF 4

Words-per-second turned out to be the most diagnostic feature in the dataset



KEY DATA POINT *Hype/Rap mean wps: 2.469 · Average of other 5 archetypes: 1.111. Hype/Rap is more than 2 standard deviations away from every other class.*

FINDING

Hype/Rap clusters at ≈2.5 words per second. Every other archetype clusters at ≈1.0 wps — and they're nearly indistinguishable from each other on this dimension.

WHY

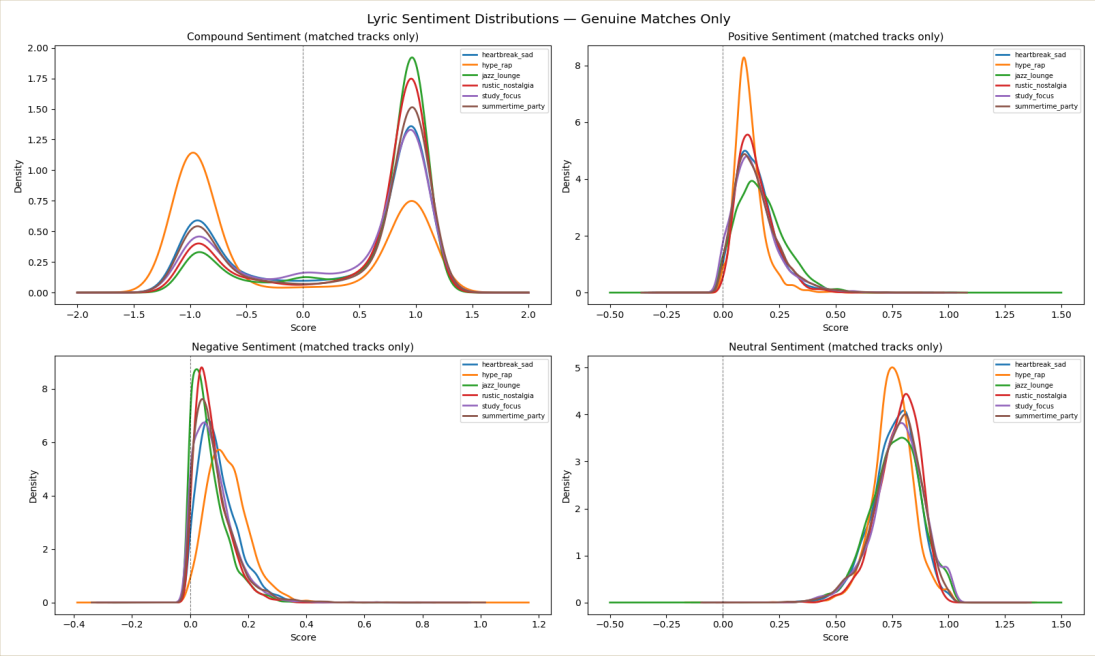
Rap is built around rapid lyrical delivery; the median rap track packs 2× the syllables of a comparable pop track over the same duration. Words-per-second is essentially a lyric-density measurement — invisible without an NLP pipeline.

WHAT IT MEANS FOR THE MODEL

This single feature ranks #1 in our final model's feature importance. It's the foundation of why Hype/Rap gets 0.84 F1 — the highest of any class — and it would never have existed without the lyric pipeline.

EDA · FINDING 3 OF 4

VADER sentiment captures lyric mood — but reveals a paradox



KEY DATA POINT *Heartbreak/Sad has +0.313 compound sentiment but a median audio valence of 0.25 — the largest contradiction between lyric and audio mood in the dataset.*

FINDING

Hype/Rap has the most negative compound sentiment (-0.187). Jazz/Lounge has the most positive (+0.573). But Heartbreak/Sad scores POSITIVE (+0.313) — exposing a real limit of lexical sentiment.

WHY

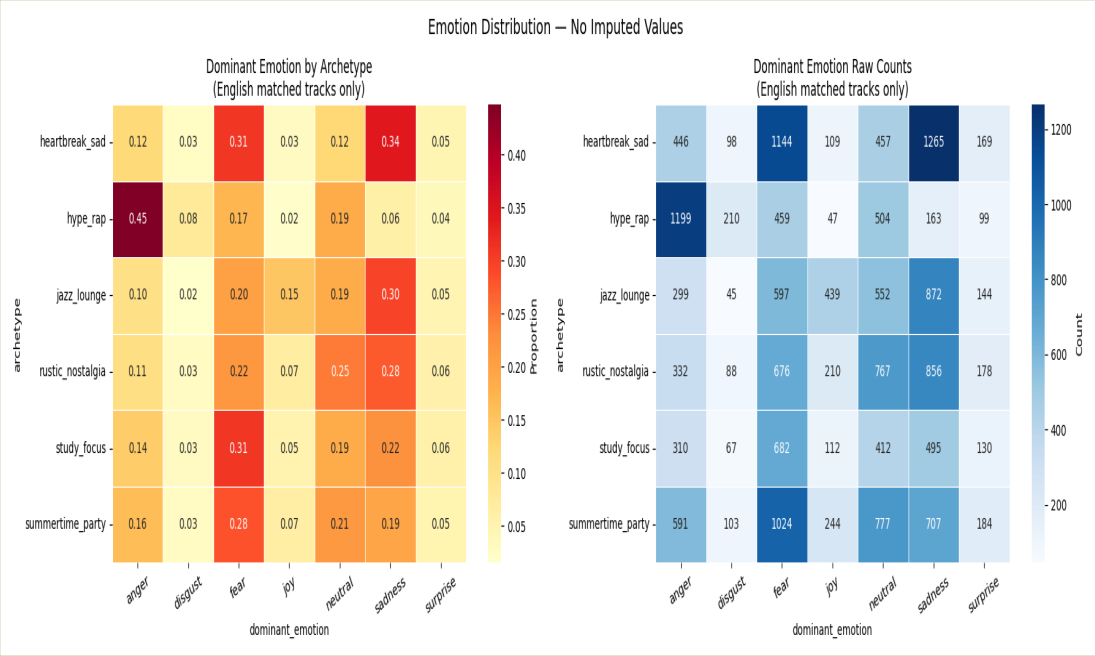
VADER scores at the word level. Heartbreak lyrics are full of "love," "miss," and "feel" — words VADER reads as positive even when the song's narrative is grief. The word "love" alone biases the score upward.

WHAT IT MEANS FOR THE MODEL

Lyric sentiment alone misses the affective truth of sad music. We need the j-hartmann emotion classifier (next slide) and audio valence (low for Heartbreak) to compensate. This is the single strongest argument for multimodal features in this project.

EDA · FINDING 4 OF 4

The j-hartmann emotion classifier explains general song writing patterns



KEY DATA POINT *Summertime/Party tracks have only 8% joy in their lyrics — modern "party" lyrics are far more melancholic than the audio suggests. We'd never have known this without the NLP pipeline.*

FINDING

Hype/Rap is dominated by anger (32%). However, the other archetypes tend to write about joy/sadness/neutral very commonly, which is likely just a tendency about how people write songs. People tend to write songs about the same emotions.

WHY

j-hartmann is a transformer fine-tuned for seven-class emotion. It reads narrative context, not just words — which is why it captures "sad" in Heartbreak even when VADER labeled it positive.

WHAT IT MEANS FOR THE MODEL

Emotion features capture nuance that lexical sentiment misses, and they land in the top 10 of feature importance. Together they let the model build per-archetype emotional profiles — not just lexical or audio ones.

MODELING · COMPARISON

Seven models on the same 25,689-track training set

Model	Family	Train acc	Test acc	Macro F1	Tuning
Logistic Regression	Linear	58.2%	58.7%	0.588	—
Decision Tree	Tree	65.0%	60.0%	0.595	—
Decision Tree (GridSearch)	Tree	65.6%	60.9%	0.604	GridSearch · 48 candidates
Random Forest	Ensemble	98.3%	68.5%	0.681	—
Random Forest (Optuna)	Ensemble	97.2%	68.6%	0.681	Optuna TPE · 60 trials
LightGBM	Boosting	98.6%	69.3%	0.689	—
LightGBM (Optuna) — BEST	**Boosting**	**96.3%**	**69.8%**	**0.695**	**Optuna TPE · 40 trials**

WHY THE NUMBERS MOVE THE WAY THEY DO

- **Linear → tree (+0.01 F1) —** Decision trees barely beat Logistic Regression. The signal isn't very linearly separable, but a single tree is too weak to exploit it.
- **Tree → ensemble (+0.09 F1) —** Bagging (RF) and boosting (LightGBM) both unlock non-linear feature interactions. The 9-point F1 jump is the headline modeling result.
- **RF → tuned LightGBM (+0.014 F1) —** Tuning RF for 60 trials added basically nothing — RF was already saturated. Boosting is just better-suited to this feature space.

BEST MODEL · PER-CLASS BREAKDOWN

Tuned LightGBM: where it works, and where it struggles

69.8%	0.695	0.686	96.3%	Δ=0.27	40
Test accuracy	Test macro F1	CV macro F1	Train accuracy	Train–test gap	Optuna trials

Archetype	Precision	Recall	F1	Support
Hype / Rap	0.82	0.87	0.84	697
Jazz / Lounge	0.81	0.70	0.75	1,550
Study / Focus	0.82	0.70	0.75	1,621
Rustic / Nostalgia	0.62	0.70	0.66	718
Heartbreak / Sad	0.54	0.64	0.59	898
Summertime / Party	0.54	0.62	0.58	939

READING THIS TABLE

- **Hype/Rap (0.84 F1)** — the easy class. words_per_second + speechiness almost solve it alone.
- **Study/Focus & Jazz/Lounge (0.75 F1)** — high precision (0.81+); model is conservative but accurate when it commits.
- **Heartbreak/Sad & Party (0.58 F1)** — the hard pair. Both vocal pop in similar tempo + valence space.
- **Macro F1 = 0.695** — every class above 0.55. No catastrophic failures.

FEATURE ABLATION

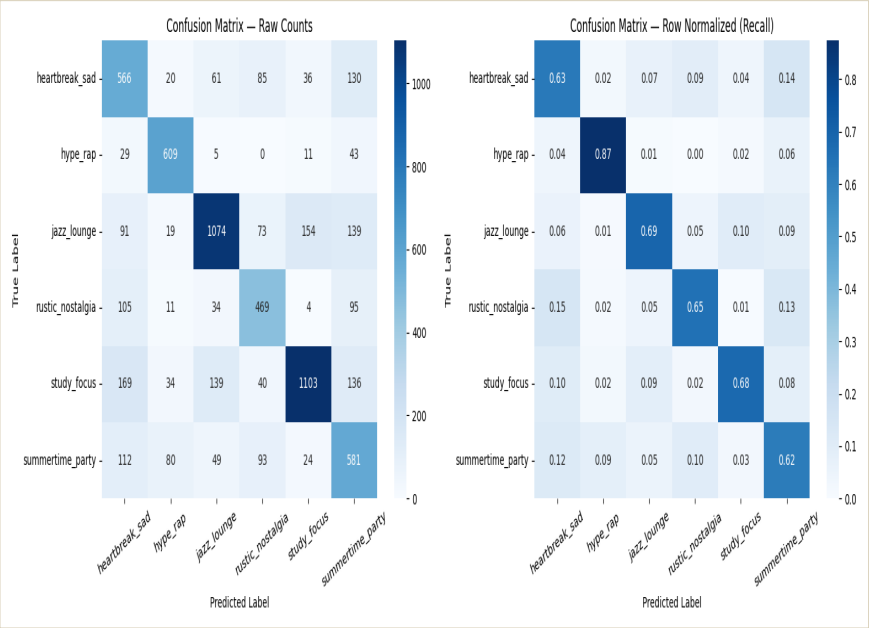
How much does each feature group actually contribute?

+0.10 macro F1 from combining audio with lyrics — orthogonal signals, not redundant ones.

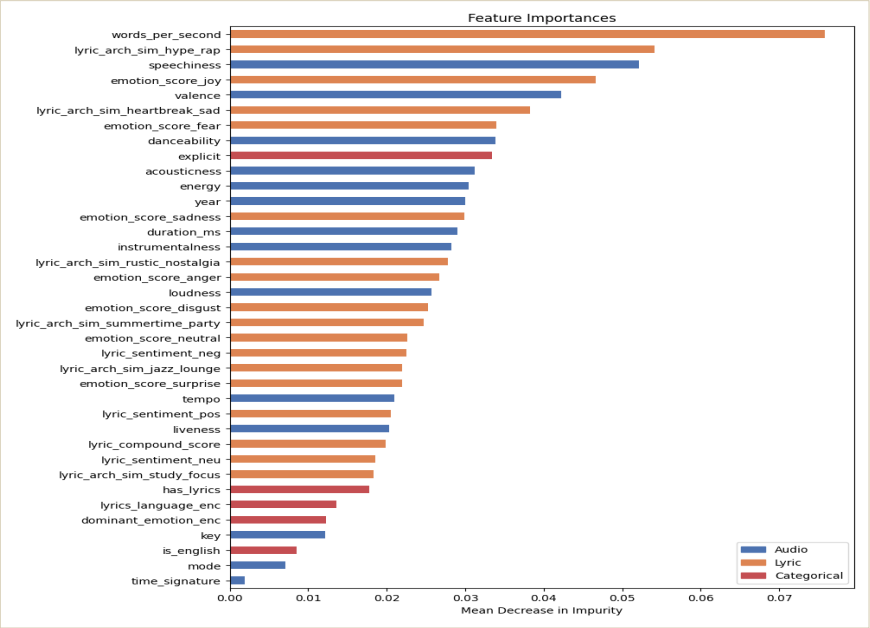
If audio and lyric features encoded the same information, combining them would give a marginal gain. A 10-point F1 jump means each modality is carrying signal the other doesn't have. This is the strongest piece of evidence in the project that NLP belongs in modern music classifiers.

DIAGNOSTICS

Where the errors concentrate and which features matter most



CONFUSION MATRIX · test set, RF baseline



FEATURE IMPORTANCE · mean decrease in impurity

Hype/Rap is closed (4% bleed). Heartbreak and Party 14% and Heartbreak and Rustic 15% bleed — most of the model's errors. *Lyric features dominate the top half of importance — NLP earned its keep.*

IMPLICATIONS & INSIGHTS

What this work says about music recommendation

Cold-start coverage impact

Vibe Check tags any new release the moment its audio and lyrics exist — which is the day it's uploaded. That's a real gap in current systems: Spotify Daily Mix and Apple For-You both need weeks of stream data before a new track surfaces in mood-based discovery.

Multimodal lift is real and quantifiable

A +0.10 macro F1 jump from combining audio with lyrics is the kind of evidence supporting lyrical data value. Modern classifiers should not skip the lyric channel, even with a 71% match rate and imputed nulls, lyric features land in the top half of importance.

Hard classes are interpretable, not random

Heartbreak and Party confusion isn't model failure, it's the data telling us emotional pop occupies overlapping audio space. Future versions should probably allow soft, multi-label predictions rather than forcing one of six.

Production economics are favorable

Tuned LightGBM trains on a laptop and predicts in seconds. No GPU, no fine-tuning, model file under 50 MB. At streaming scale this comes out to fractions of a cent per million predictions.

LIMITATIONS & FUTURE WORK

Challenges/Limitations and Future Work

CHALLENGES WE FACED

- Accurate class data collection was a challenge because people have different understandings of the classes, and key word search likely can’t make up for that
- Lyric coverage uneven — 71% match rate overall, but Study/Focus and Jazz/Lounge are mostly instrumental.
- Class imbalance addressed with weights — but Hype/Rap recall ceiling is still capped at 0.87 by training-set size.
- Sentence-transformer similarity didn't recover the right archetype for Jazz/Lounge or Rustic/Nostalgia — known weakness.

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LIMITATIONS OF THE METHOD

- Single-label classification — Bon Iver's "Skinny Love" is heartbreak AND rustic, but we force a choice.
- English-centric NLP — the 30% of non-English tracks get reduced lyric signal.
- Spotify-feature dependence — features don't transfer to Apple Music or YouTube without re-derivation.
- Year drift — features have shifted over time (loudness war, etc.); model has implicit recency bias.

FUTURE WORK

- Multi-label / soft archetype scores with calibrated probabilities (Platt scaling).
- Multilingual lyric model — XLM-R or Bloom fine-tune for non-English tracks.
- Active learning loop with human-in-the-loop playlist editor labels.
- Audio embeddings (CLAP, OpenL3) as a richer alternative to Spotify's hand-crafted features.

Thank *you*

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